



Instructor: Date:

Learner: Regional Branch:

Program: Center:

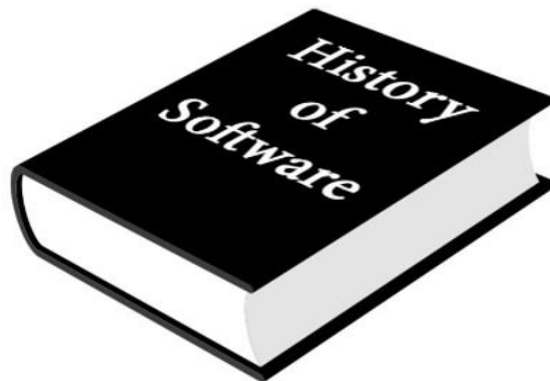
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Let's Start!

Discovering How Software Engineering Started.



Learning Outcome:

Describe common activities at work to a using short sentences in familiar spoken and written situations.

Make the result report from the analysis of the information system, based on the customer requirements, following established rules and protocols.

Ask your partner:

- What facts do you know about the history of computer programming?
- Why do you think software development is constantly evolving?

Can Do Statements:

- I can read about the history of software.
- I can summarize the information in a short paragraph.
- I can understand important information related to software engineering.





Let's Explore!

Blogging About Software.



Instructions: Read the text below.

As you know, the intention of this blog is to post information about technology, computing systems, and programming languages. Today, I want to focus on the history of software and some related information to software engineering.

Programming devices existed since very early times but it wasn't until the mid nineteenth century that Ada Lovelace invented the first computer program. Years later, Herman Hollerith introduced the concept of storing data and IBM launched the first electronic computer in the late 1940's.

But, when did software engineering truly emerge? At the beginning of the XX century, there were many people dedicated to create and improve computer systems, however, this field didn't emerge as a profession until the 60's. From that time on, software developers aimed and still aim to guarantee the stability, speed, usability, testability, readability, size, cost, and security of computer programs.

The evolution of this field is remarkable in different areas. First of all, processes, as we may know, are now fundamental in software engineering and they led to all the progress that we can see now in programming languages, data storage devices or computer terminals. On another note, we can notice how the cost of software and hardware changed with the evolution of computer programming. By the late 60's, it was really expensive to afford mainframes and support staff, but the boom of computers made these machines cheaper and more companies are now able to support larger software projects.

Another interesting fact is the rise of the internet, thanks to the creation of the web browser in the early 90's. This led to a fast growth in the demand for international information, and improvements in the email systems, network connections, and media display and storage were necessary.

Nowadays, all the rapid changes in technology, the current tendency in software development, is creating programs and apps for smaller companies, and programmers are constantly developing agile lightweight methodologies, and alternative approaches to software calculations and information storage.



Let's Practice!

Talking About the History of Software.

Instructions: Write the words to correctly complete the sentences. Use words from the box below.

dedicated / didn't emerge / did / were

But, when _____ software engineering truly emerge? At the beginning of the XX century, there _____ many people _____ to create and improve computer systems, however, this field _____ as a profession until the 60's.



Let's Practice!

The evolution of this software engineering is remarkable in different areas.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. Software developers aimed and...
2. Processes are now fundamental in software engineering and they led to...
3. The boom of computers made these machines cheaper...

COLUMN B

- a. and more companies are now able to support larger software projects.
- b. all the progress in programming languages, data storage devices, and computer terminals.
- c. still aim to guarantee the quality requirements of computer programs.



Extension Activity

Creating a Brochure About the History of Software.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the history of software.
- Identify 4 main facts about the history of software.
- Create a brochure with the information you found.
- Write a short paragraph to explain each fact in your brochure.
- Share your brochure with a partner.

Self-Assessment

Yes

No

- | | | |
|---|-----------------------|-----------------------|
| 1. I look for more information about history of software. | ☐ | ☐ |
| 2. I identify 4 main facts about the history of software. | ☐ | ☐ |
| 3. I create a brochure with the information I found. | ☐ | ☐ |
| 4. I write a short paragraph to explain each fact in my brochure. | ☐ | ☐ |
| 5. I share my brochure with a partner. | ☐ | ☐ |



Target Words

- **rise** (noun): the emergence of something
e.g. Another interesting fact is the rise of the internet, thanks to the creation of the web browser in the early 90's.
- **readability** (noun): easy and available to be used by a computer
e.g. From that time on, software developers aimed and still aim to guarantee the stability, speed, usability, testability, readability, size, cost, and security of computer programs.
- **launch** (verb): to introduce something
e.g. IBM launched the first electronic computer in the late 1940's.
- **lead** (verb): to be a means to produce something
e.g. They led to all the progress that we can see now in programming languages, data storage devices or computer terminals.
- **lightweight** (adjective): something that weights only a little or less than average
e.g. Programmers are constantly developing agile, lightweight methodologies, and alternative approaches.



Learning Points

Simple Past Review.

You use the **simple past** tense to talk about situations that started and finished at a specific time in the past.

Affirmative

To make affirmative sentence in **simple past**, you should use verbs in past form:

Subject + **verb in past form** + complement.

Examples:

*Ada Lovelace **invented** the first computer program in 1843.*

*Herman Hollerith **introduced** the concept of storing data in the 1880's.*

*Data storage devices **became** less expensive by the 1960's.*

Verbs in past are classified in **regular** and **irregular**:

The past form of **regular verbs** is simple, you just need to add **-d**, **-ed**, or **-ied** at the end of the verb.

Examples:

*create → **created***

*invent → **invented***

*study → **studied***

Irregular verbs can change in different forms, you need to practice and memorize them.

Examples:

*keep → **kept***

*think → **thought***

*become → **became***

Negative

To make negative statements in **past simple** with the verb **to be**, you use the following structure:

Subject + **wasn't/weren't** + complement.

Examples:

*Ada Lovelace **wasn't** the first person that invented a computer program.*

*Computer programs **weren't** common in the XIX century.*





Learning Points

When you make negative statements with other verbs; you don't have to use the past form of the verb but the infinitive form. Also, you need to include the auxiliary **did** in negative before the main verb:

Subject + **didn't** + **verb in infinitive** + complement.

Examples:

*Herman Hollerith **didn't introduce** the concept of storing data in the 1880's.*

*Data storage devices **didn't become** less expensive by the 1960's.*

Interrogative

To make questions in **past simple** with the verb **to be**, you use the following structure:

Was/were + subject + complement?

Examples:

Was *Ada Lovelace the first person that invented a computer program?*

Were *computer programs common in the XIX century?*

When you ask questions with other verbs; you don't have to use the past form of the verb but the infinitive form. Also, you need to include the auxiliary **did** at the beginning of the question:

Did + subject + **verb in infinitive** + complement?

Examples:

Did *Herman Hollerith **introduce** the concept of storing data in the 1880's?*

Did *data storage devices **become** less expensive by the 1960's?*

You can also use WH words in questions when you are asking for specific information:

Who was *Adam Lovelace?*

Why were *computers programs more common in the XX century?*

Why did *data storage devices become less expensive by the 1960's?*

Examples

- *Programming devices existed since very early times but it wasn't until the mid-nineteenth century that Ada Lovelace invented the first computer program.*
- *But, when did software engineering truly emerge? At the beginning of the XX century, there were many people dedicated to create and improve computer systems, however, this field didn't emerge as a profession until the 60's.*





Learning Points

Understanding a Paragraph.

A paragraph is a group of sentences connected by the logic of a single idea. This way, the sentences within the paragraph develop just one main idea. A paragraph should have three essential parts: a topic sentence, some supporting sentences, and a concluding sentence.

Take a look at the following chart:

Topic Sentence	• It is the most general sentence in the paragraph; all the following sentences are more specific. • It states the purpose of the paragraph for the reader. • It is the most important sentence in the paragraph. It covers all of the supporting ideas in the paragraph.
Supporting Sentences	• These sentences are used to support and explain your supporting sentences in a more detailed way. • For constructing the supporting sentence, you can ask questions such as why, how, or when in order to develop the information of the topic sentence. • You can also use facts and statistics.
Closing Sentences	• This sentence is included in the last part of your paragraph. • In a concluding sentence, you should summarize the topic sentence and the main points of your paragraph. • Do not use the same words you used in the topic sentence.

